

Artificial Intelligence in Military: An Indian Perspective

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Abstract. Artificial Intelligence (AI) is widely acknowledged as a transformative technology with profound implications for national security and military operations. India, with its growing IT sector and strong engineering talent pool, is actively working to integrate AI capabilities into its defence infrastructure. This paper examines the current state of AI adoption in the Indian military, reviews key robotics and autonomous systems developed by Indian defence organisations, and analyses the strategic challenges and opportunities India faces as it seeks to develop a credible AI-based defence capability. The paper draws on publicly available policy documents, defence reports, and academic literature to present a structured overview of India's AI ambitions in the military domain.

Keywords: Artificial Intelligence, Indian Military, Defence Technology, Autonomous Systems, Robotics, DRDO, National Security, Machine Learning

1 Introduction

Artificial Intelligence (AI) has the potential to bring significant transformations in the way both civilian activities and military operations are conducted. AI is fundamentally a dual-use technology: while it can fuel technology-driven economic growth, it simultaneously holds the potential to provide military superiority to nations that deploy it effectively [1].

India possesses a strong information technology industry and a substantial pool of engineering talent — structural advantages that position it well for AI development. The Ministry of Defence has already initiated the process of preparing Indian defence forces for the use of AI, establishing task forces and publishing strategic frameworks to guide the integration of AI capabilities into the armed forces [2].

The key objectives examined in this paper are as follows:

- Assessing the current level of AI and machine learning development in India, with specific focus on defence applications
- Reviewing recommendations for making India a significant AI power across aviation, naval, land systems, cyber, and biological warfare domains
- Examining policy and institutional frameworks required to regulate and encourage robust AI-based defence technologies
- Analysing opportunities for collaboration between the defence sector and private industry, particularly technology start-ups [3]

2 Strategic Context: What Kind of AI Does India Need?

A fundamental question facing Indian defence planners is the appropriate level of autonomy to extend to AI systems on the battlefield. Should India develop fully autonomous drones capable of engaging adversary aircraft, or focus on supervised autonomous systems for border surveillance and logistics support? How much decision-making authority should be delegated to machines in high-stakes military contexts?

These questions carry significant ethical, legal, and strategic dimensions. For a middle-income country like India, which must balance defence investment against pressing public welfare needs, a clearly articulated and prioritised AI vision is essential. The lack of critical computing infrastructure remains one of the biggest impediments to AI adoption, as these systems require robust hardware and large-scale data infrastructure [4].

2.1 India's Position in the Global AI Race

The global AI arms race is currently dominated by the United States, China, and Russia — nations that have made substantial investments in military AI. AI, as a dual-use technology, has opened the possibility for middle powers with strong civilian AI sectors to compete in the defence domain. India's position is therefore both promising and precarious.

In January 2019, Army Chief General Bipin Rawat stated that India would be too late if its armed forces did not embrace AI soon enough. This statement underscores the urgency felt within Indian defence leadership and has accelerated institutional efforts to embed AI into defence planning and procurement processes.

3 AI and Robotics Systems in the Indian Military

The Centre for Artificial Intelligence and Robotics (CAIR), operating under the Defence Research and Development Organisation (DRDO), has developed a range of AI-enabled robotic systems. These systems demonstrate India's growing indigenous capability in defence robotics and autonomous systems [5].

3.1 Daksh — Remotely Operated Vehicle

Daksh is an electrically powered, remotely controlled robot designed for the location, handling, and safe destruction of hazardous objects. Developed entirely by DRDO, it is battery-operated and travels on wheels. Ninety per cent of its components are indigenously manufactured. Key capabilities include [6]:

- Navigation of staircases, steep slopes, narrow corridors, and vehicle access points
- Robotic arm for lifting and scanning suspect objects using a portable X-Ray device
- Water jet disrupter for bomb defusal operations
- Integrated shotgun for breaching locked doors
- Remote control capability over 500 metres line-of-sight or within buildings

3.2 Snake Robots and Hexa-Bots

CAIR has also developed snake robots capable of covert surveillance in terrain inaccessible to conventional systems. Equipped with cameras that transmit video feeds to field commanders, these robots can navigate confined spaces where human soldiers or wheeled robots cannot operate. Hexa-bots — six-legged robotic platforms — offer additional terrain adaptability for complex and uneven environments.

3.3 Border Patrol Robots

Bharat Electronics Limited (BEL) has developed border patrol robots equipped with sensors and communication systems to interface with central control stations. These robots are designed to reduce risk to human security personnel deployed in dangerous border surveillance roles, and can operate in high-threat environments including blast and contamination zones.

4 Key Challenges and Policy Implications

Despite the progress noted above, India faces several structural challenges in scaling its military AI programme:

- Infrastructure deficit: AI systems require robust computational hardware and large-scale data repositories. India's current capacity in both areas remains insufficient for large-scale military AI deployment.
- Talent retention: India produces significant AI engineering talent, but retention within the defence sector is challenging given competitive private sector compensation packages.
- Ethical and legal frameworks: The deployment of autonomous weapons systems raises unresolved questions about accountability and compliance with international humanitarian law.
- Private sector integration: Most AI development is occurring in the private sector. Effective collaboration between DRDO, the armed forces, and commercial start-ups is critical but currently underutilised.
- Data availability: Military AI requires large volumes of labelled operational data. Access to such data within appropriate security classifications remains a significant bottleneck.

5 Conclusion

India stands at a critical juncture in its military AI journey. The foundational elements are in place — a strong engineering workforce, growing indigenous robotics capability, and an engaged policy environment. However, translating these assets into operational AI superiority will require sustained investment, institutional reform, and a clearly articulated national AI defence strategy.

The window of opportunity is narrowing as global AI development accelerates. India's ability to define and execute a pragmatic, well-resourced AI programme — one that balances strategic ambition with fiscal reality — will be a defining factor in its future security posture. Closer collaboration between government, academia, and private industry will be essential to achieving this goal.

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